

Automation or Rising Wealth?

Results of a horse race between technological push and demand pull in the growth of US low-skill service employment, 1980–2005

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This document reports the results produced by the accompanying Python code (notebook TrackB_AutorDorn.ipynb / script trackB_analysis.py) and comments on them. It is the Track B “results obtained with commentary” deliverable.

1. Dataset used

We work with the public replication corpus of Autor & Dorn (2013), “*The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market*” (AER 103(5): 1553–1597), deposited in the AEA Data and Code Repository / openICPSR, project 112652 (DOI 10.3886/E112652V1). The unit of observation is the US **commuting zone** (CZ), a local labour market; the panel covers 722 CZs spanning the entire mainland US, observed at the Census years 1950, 1970, 1980, 1990, 2000 and the 2005 ACS. We use three files: *workfile2012.dta* (the CZ panel: routine employment share, noncollege service-employment change, wages, controls and the 1950 industry-mix instrument), *RTI-bypctile-1980.dta* (routine occupation share by skill percentile) and *dwg-byperc-1980-2005-czall.dta* (observed and counterfactual wage growth by percentile). The analytical sample stacks the three decadal differences 1980–2005, giving $722 \times 3 = 2,166$ CZ-period observations. The routine share has a population-weighted mean of 0.33 in 1980 and an 80/20 range of 0.072.

2. Descriptive results

Figure 1 establishes the phenomenon. Panel A reproduces the wage U-shape: between 1980 and 2005 real wage growth was strong at the top of the skill distribution, weakest around the median, and modest but positive at the bottom. The dashed counterfactual — in which service-occupation wage growth is set to zero — collapses at the lower tail, showing that the upward twist of the bottom of the distribution is *driven by service occupations*. Panel B locates routine work along the same skill axis: it peaks in the middle of the distribution, exactly where wage growth is weakest. The middle hollowed out where routine, automatable jobs were concentrated.

Output 1 — Wage polarization and the location of routine jobs

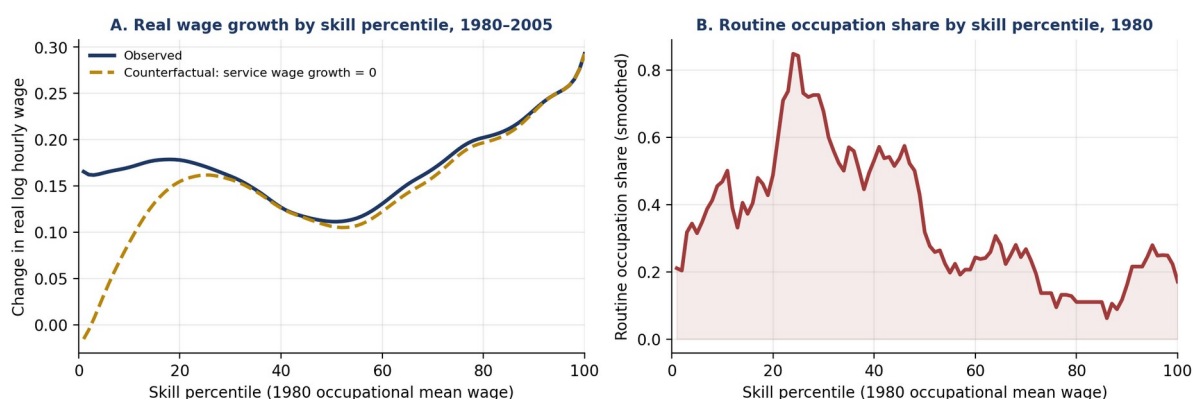


Figure 1 — Output 1. Wage growth by skill percentile, observed vs counterfactual (Panel A), and routine occupation share by skill percentile (Panel B). Source: *dwg-byperc* and *RTI-bypctile* files (Autor & Dorn, 2013).

Figure 2 turns to the employment side and introduces the cross-CZ variation our analysis exploits. Classifying CZs as “high-routine” or “low-routine” by whether their 1980 routine share lies above or below the population-weighted national mean, we plot the average decadal change in noncollege employment by major occupation group. In high-routine CZs, service employment rises *more*, while routine occupations (clerical/retail, production, machine operators) fall *more* than in low-routine CZs. Labour reallocated out of the automatable middle and into in-person services precisely where there was more routine work to automate.

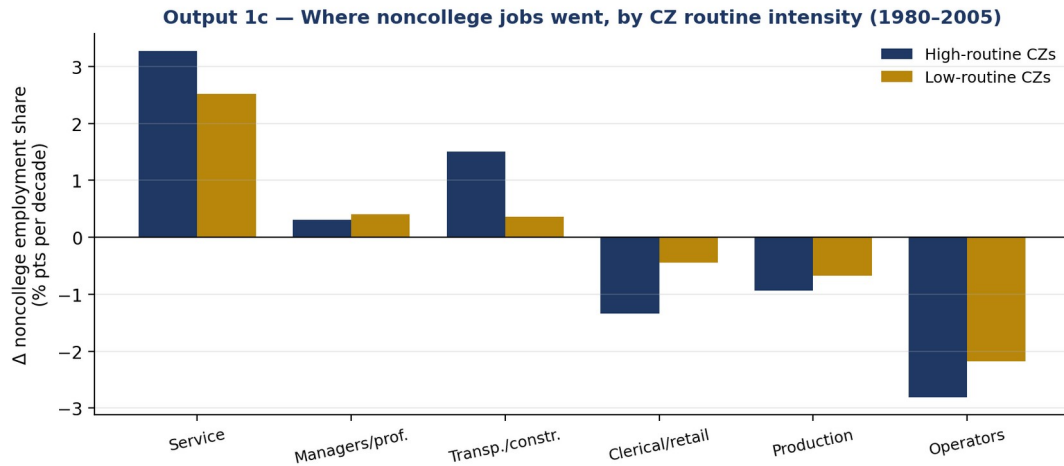


Figure 2 — Output 1c. Change in noncollege employment share by occupation group, high- vs low-routine commuting zones, 1980–2005.
Source: *workfile2012.dta*.

3. Analytical results — the horse race

We regress the decadal change in the noncollege service-employment share on the two candidate drivers and let them compete. Table 1 reports a staircase: the routine share alone (1), top-wage growth alone (2), both together (3), both plus college labour supply and offshorability (4), and a 2SLS estimate (5) that instruments the routine share with the predetermined 1950 industry-mix prediction and adds the full Autor & Dorn control set with state and time fixed effects. Columns (1)–(4) are population-weighted OLS; all standard errors are clustered on state.

Table 1 — Horse race: determinants of noncollege service-employment growth, 1980–2005

Dependent: Δ service share	(1) RSH	(2) Demand	(3) Horse race	(4) +controls	(5) 2SLS
<i>RSH</i> (routine share)	0.124***	—	0.096***	0.173***	0.149***
Δ top wage (<i>P90</i>)	—	0.048***	0.037***	0.012	—
Δ college labour supply	—	—	—	-0.110***	—
<i>Offshorability</i>	—	—	—	-0.005	—
State & time FE	No	No	No	No	Yes
Instrument (1950 mix)	No	No	No	No	Yes
<i>N</i>	2,166	2,166	2,166	2,166	2,166

Weighted by start-of-period CZ population share; standard errors clustered on state. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The column-(5) first-stage F-statistic is 174, far above conventional weak-instrument thresholds, and its coefficient (0.149) reproduces Autor & Dorn's headline 2SLS estimate.

The reading is clear. The routine share is positive and highly significant on its own (col 1), **survives** the head-to-head with top-wage growth (col 3), the addition of demand-side controls (col 4), and full instrumentation (col 5). Top-wage growth, by contrast, is significant in isolation (col 2) but **attenuates** once the routine share is present (col 3) and becomes indistinguishable from zero with controls (col 4). College labour-supply growth even enters *negatively* — the opposite of what a pure demand-pull/substitution story predicts. On this dataset, the technological-push channel accounts for the cross-CZ growth of service employment; the demand-pull proxies do not survive conditioning.

Figure 3 shows the same contest visually, on the 722-CZ cross-section. Both raw slopes are positive because the two mechanisms are *entangled* across space: routine-intensive CZs also enjoyed faster top-wage growth (the unconditional correlation is ≈ 0.29). That entanglement is exactly why the scatter alone cannot adjudicate, and why the regression in Table 1 — which holds one mechanism fixed while varying the other and instruments the routine share with predetermined 1950 industry structure — is needed. Once that is done, only the technological-push channel survives.

Output 2 — The horse race across 722 commuting zones

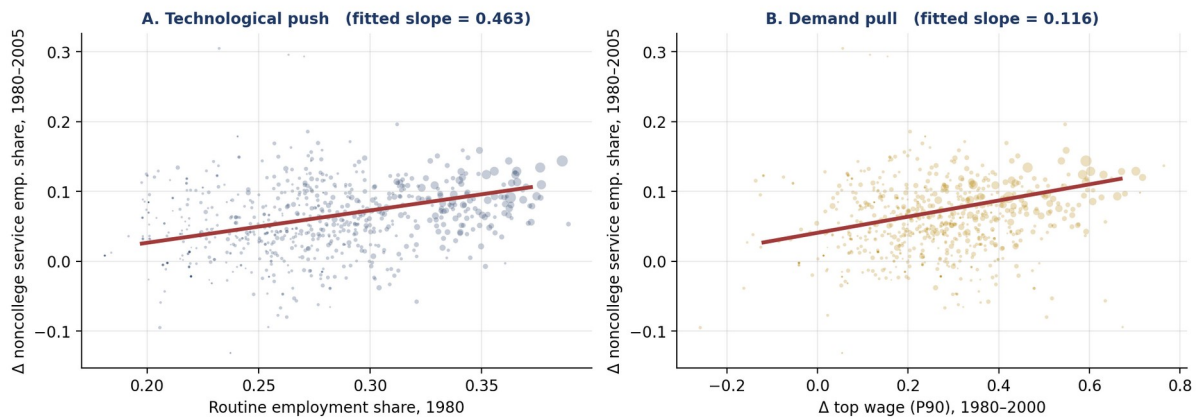


Figure 3 — Output 2. Change in noncollege service-employment share (1980–2005) against the 1980 routine share (Panel A) and against top-wage growth (Panel B), 722 commuting zones, population-weighted fit. Source: *workfile2012.dta*.

4. Reading the results against the research question

Our question was: placed in the same specification on the same data, which mechanism accounts for the cross-CZ growth of low-skill service employment? The answer the corpus gives is **technological push**. The descriptive evidence (Figures 1–2) is consistent with both stories, as expected — the polarization fact does not by itself identify a cause. The analytical evidence (Table 1, Figure 3) breaks the tie: the routine-task measure is robust across every specification and reproduces the published instrumented estimate, while the demand-pull proxies fade under conditioning. This does not deny that local demand matters anywhere; it says that, at the level of US local labour markets and over 1980–2005, the automation of routine tasks is the dominant driver, and that the demand-pull relationship visible in raw scatters reflects the spatial co-movement of the two forces rather than an independent service-demand channel.

5. Limitations and reproducibility

Wages are deflated by the national PCE index, so cross-CZ comparisons are nominal; the routine index is fixed at 1980 task content and does not capture within-occupation task drift; the panel ends at the 2005 ACS; and the design is not built to recover a causal local multiplier in Moretti’s sense (that would require a tradable-employment shift-share instrument). The high/low-routine split in Figure 2 is computed at the occupation-group level rather than by skill percentile, because the percentile-level files are aggregated across all commuting zones and do not support a routine-group split by percentile. These are relative-adjudication results, not a universal rejection of demand effects.

Reproducibility. All results above are produced end-to-end by the accompanying code after `pip install -r requirements.txt`. The code uses no hard-coded absolute paths (it searches for the three `.dta` files), organises the work into functions, and writes every figure and the results table to an `outputs/` folder. Required packages: `pandas`, `numpy`, `pyreadstat`, `matplotlib`, `statsmodels`, `linearmodels`, `scipy` (full list in `requirements.txt`). No external vocabularies or downloads are needed beyond the three `.dta` files from openICPSR project 112652.

Acknowledgement of AI use. We used a large-language-model assistant (Claude) to inspect the `.dta` files, sanity-check summary statistics and candidate specifications, and to refine prose and plotting code. All modelling choices and interpretations are our own, and we can explain every line of the submitted code.